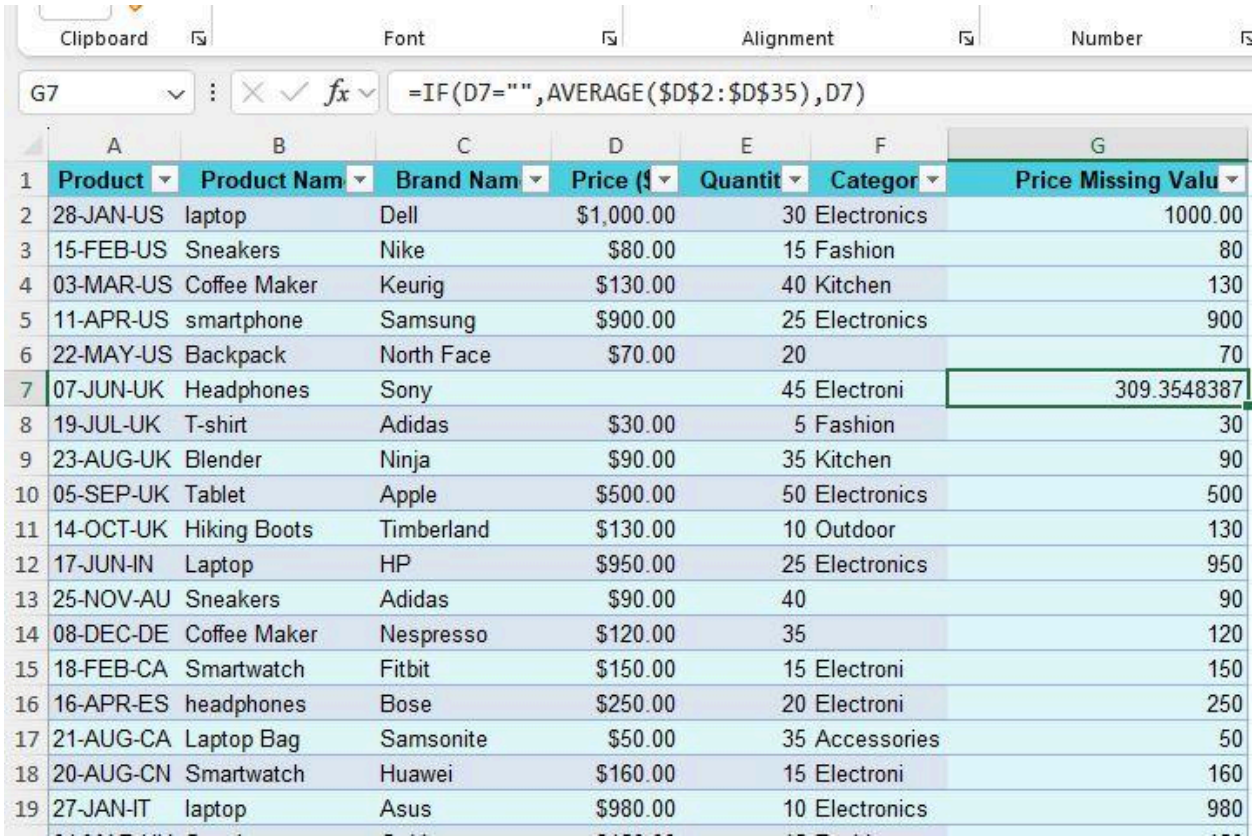
Data Analytics (DA) Module -1 - Excel, Excel Assignment 2 – Data Exploration

1. Handling Missing Values:
● Check for missing values in the 'Price' column. How would you handle products with missing price information?



	A	B	C	D	E	F	G
1	Product	Product Nam	Brand Nam	Price ($	Quantit	Categor	Price Missing Valu
2	28-JAN-US	laptop	Dell	$1,000.00	30	Electronics	1000.00
3	15-FEB-US	Sneakers	Nike	$80.00	15	Fashion	80
4	03-MAR-US	Coffee Maker	Keurig	$130.00	40	Kitchen	130
5	11-APR-US	smartphone	Samsung	$900.00	25	Electronics	900
6	22-MAY-US	Backpack	North Face	$70.00	20		70
7	07-JUN-UK	Headphones	Sony		45	Electroni	309.3548387
8	19-JUL-UK	T-shirt	Adidas	$30.00	5	Fashion	30
9	23-AUG-UK	Blender	Ninja	$90.00	35	Kitchen	90
10	05-SEP-UK	Tablet	Apple	$500.00	50	Electronics	500
11	14-OCT-UK	Hiking Boots	Timberland	$130.00	10	Outdoor	130
12	17-JUN-IN	Laptop	HP	$950.00	25	Electronics	950
13	25-NOV-AU	Sneakers	Adidas	$90.00	40		90
14	08-DEC-DE	Coffee Maker	Nespresso	$120.00	35		120
15	18-FEB-CA	Smartwatch	Fitbit	$150.00	15	Electroni	150
16	16-APR-ES	headphones	Bose	$250.00	20	Electroni	250
17	21-AUG-CA	Laptop Bag	Samsonite	$50.00	35	Accessories	50
18	20-AUG-CN	Smartwatch	Huawei	$160.00	15	Electroni	160
19	27-JAN-IT	laptop	Asus	$980.00	10	Electronics	980

● **If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.**



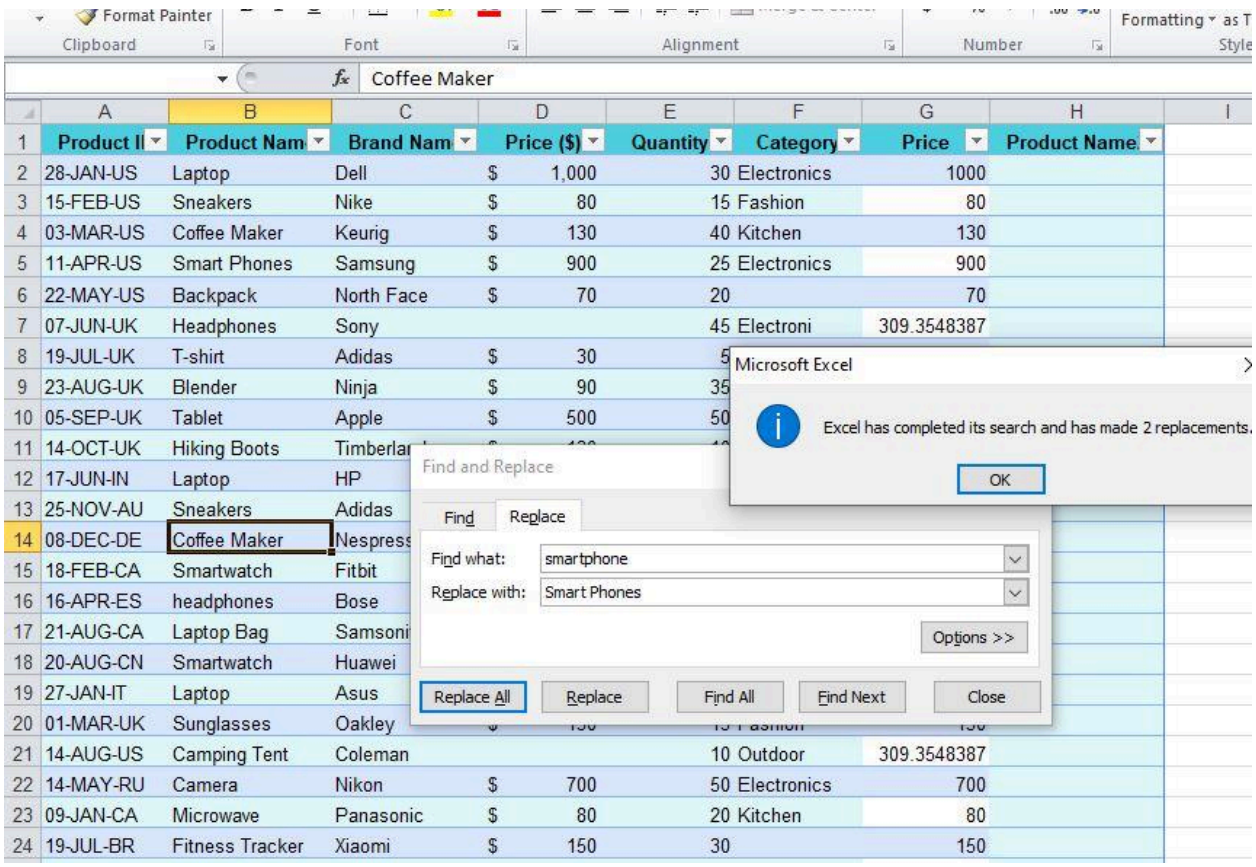
Clipboard | Font | Alignment | Number

H2 : =VLOOKUP([@[Product Name]],Table1[[#All],[Product Name]:[Category]],5,0)

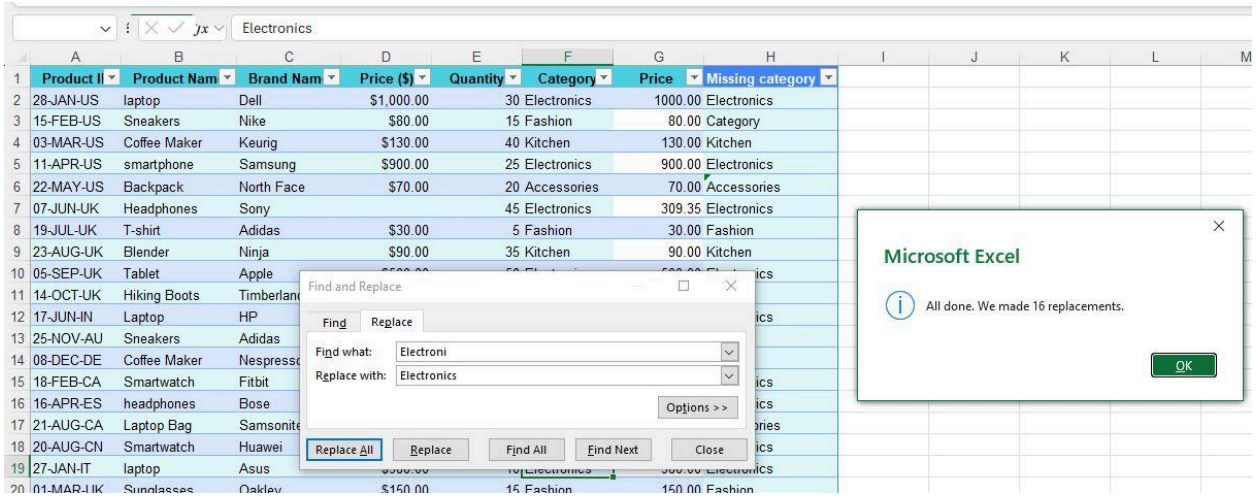
	A	B	C	D	E	F	G	H
1	Product Il	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price	Missing category
2	28-JAN-US	laptop	Dell	$1,000.00	30	Electronics	1000.00	Electronics
3	15-FEB-US	Sneakers	Nike	$80.00	15	Fashion	80.00	Category
4	03-MAR-US	Coffee Maker	Keurig	$130.00	40	Kitchen	130.00	Kitchen
5	11-APR-US	smartphone	Samsung	$900.00	25	Electronics	900.00	Electronics
6	22-MAY-US	Backpack	North Face	$70.00	20	Accessories	70.00	Accessories
7	07-JUN-UK	Headphones	Sony		45	Electroni	309.35	Electroni
8	19-JUL-UK	T-shirt	Adidas	$30.00	5	Fashion	30.00	Fashion
9	23-AUG-UK	Blender	Ninja	$90.00	35	Kitchen	90.00	Kitchen
10	05-SEP-UK	Tablet	Apple	$500.00	50	Electronics	500.00	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	$130.00	10	Outdoor	130.00	Outdoor
12	17-JUN-IN	Laptop	HP	$950.00	25	Electronics	950.00	Electronics
13	25-NOV-AU	Sneakers	Adidas	$90.00	40		90.00	Fashion
14	08-DEC-DE	Coffee Maker	Nespresso	$120.00	35		120.00	Kitchen
15	18-FEB-CA	Smartwatch	Fitbit	$150.00	15	Electroni	150.00	Electroni
16	16-APR-ES	headphones	Bose	$250.00	20	Electroni	250.00	Electroni
17	21-AUG-CA	Laptop Bag	Samsonite	$50.00	35	Accessories	50.00	Accessories
18	20-AUG-CN	Smartwatch	Huawei	$160.00	15	Electroni	160.00	Electroni
19	27-JAN-IT	laptop	Asus	$980.00	10	Electronics	980.00	Electronics
20	01-MAR-UK	Sunglasses	Oakley	$150.00	15	Fashion	150.00	Fashion
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor	309.35	Outdoor

2. Correcting Inconsistent Data:

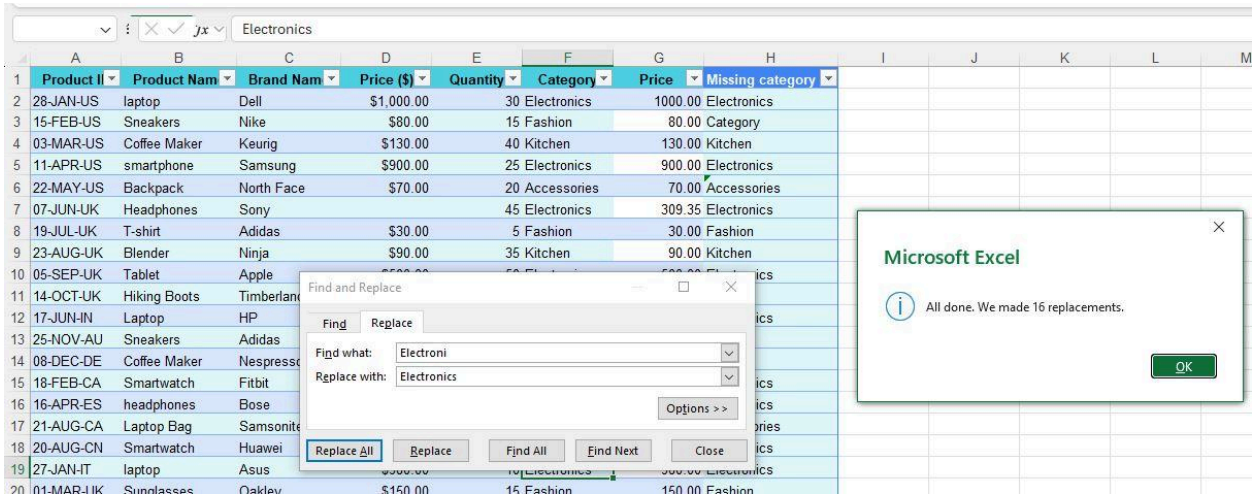
● **Identify any inconsistent text formats present in the "Product Name" column.**



- **Identify any typos present in the "Category" column.**

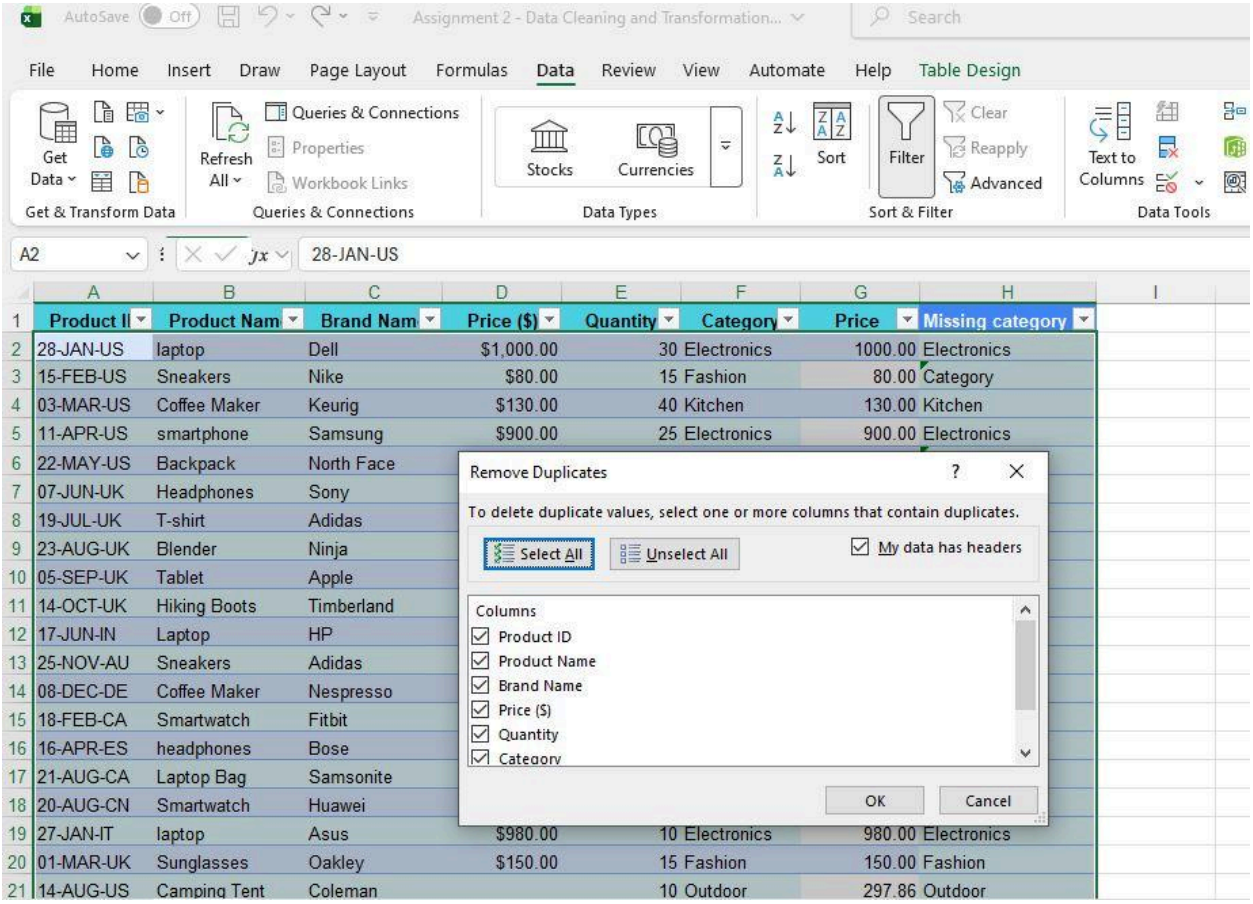


● Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.



3. Removing Duplicates:

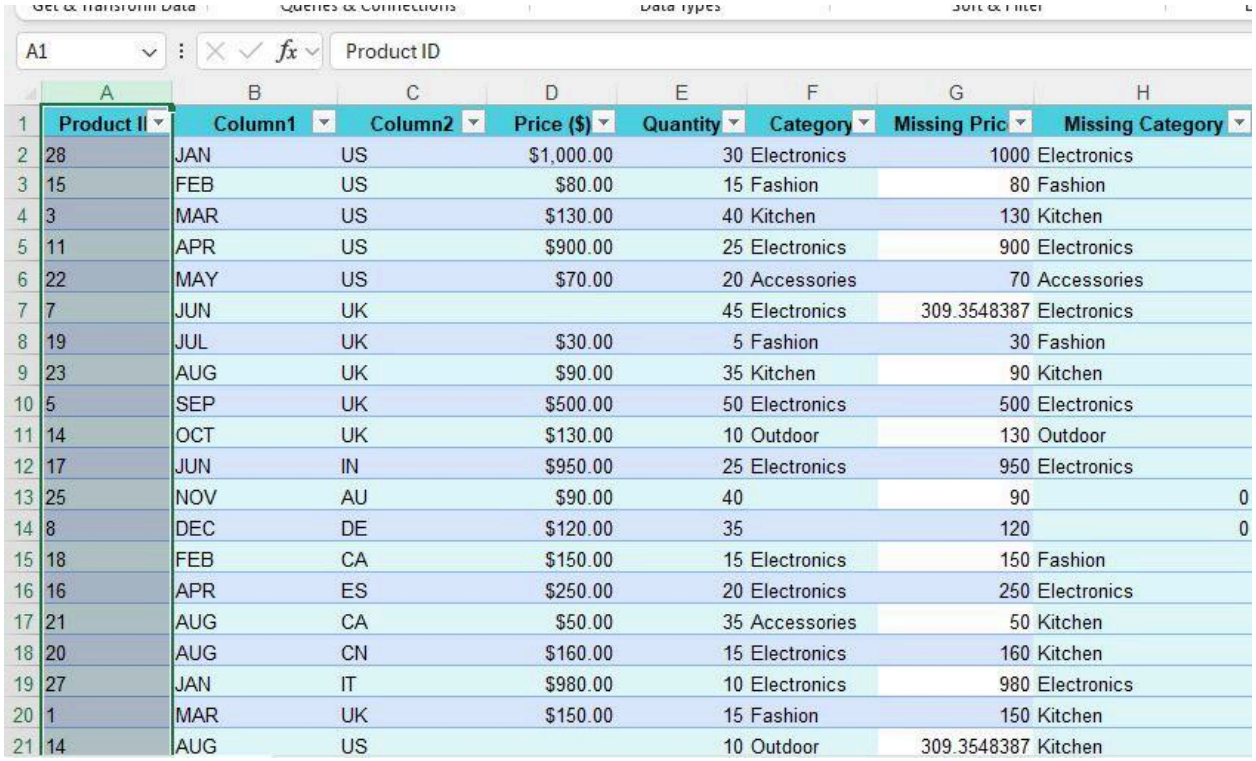
● Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.



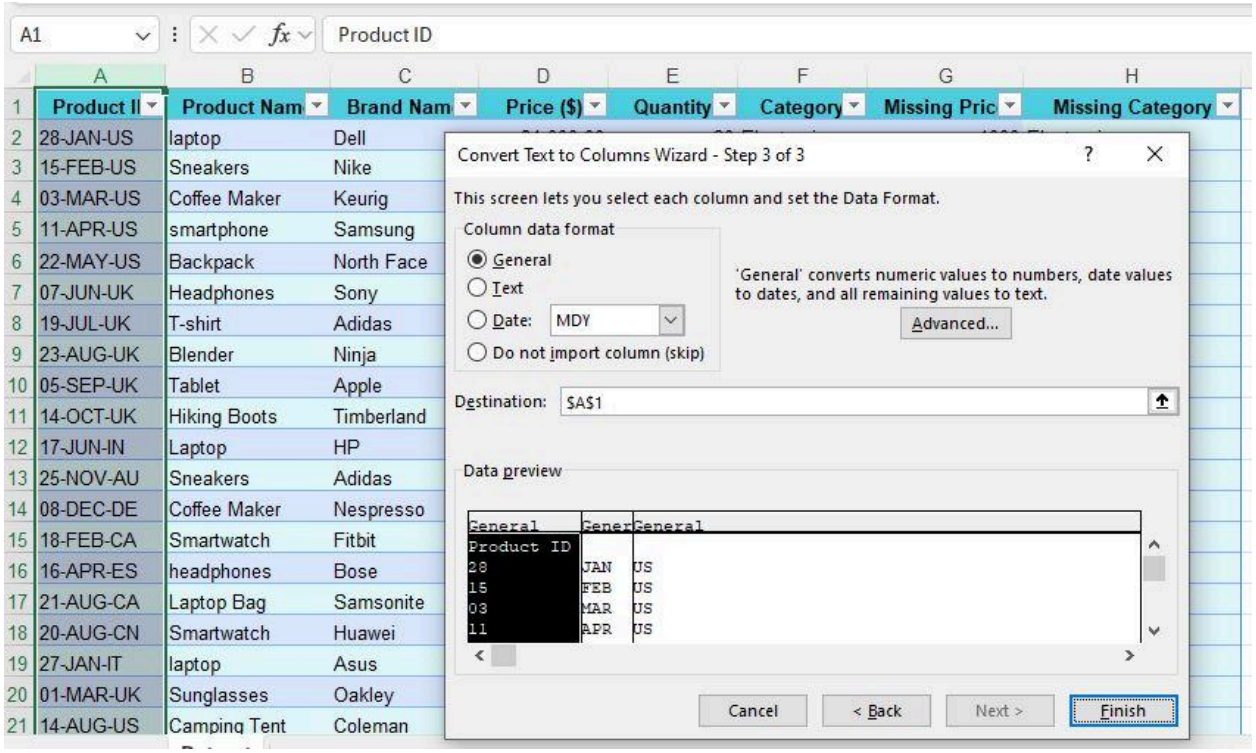
	A	B	C	D	E	F	G	H	I
1	Product Il	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price	Missing category	
2	28-JAN-US	laptop	Dell	$1,000.00	30	Electronics	1000.00	Electronics	
3	15-FEB-US	Sneakers	Nike	$80.00	15	Fashion	80.00	Category	
4	03-MAR-US	Coffee Maker	Keurig	$130.00	40	Kitchen	130.00	Kitchen	
5	11-APR-US	smartphone	Samsung	$900.00	25	Electronics	900.00	Electronics	
6	22-MAY-US	Backpack	North Face						
7	07-JUN-UK	Headphones	Sony						
8	19-JUL-UK	T-shirt	Adidas						
9	23-AUG-UK	Blender	Ninja						
10	05-SEP-UK	Tablet	Apple						
11	14-OCT-UK	Hiking Boots	Timberland						
12	17-JUN-IN	Laptop	HP						
13	25-NOV-AU	Sneakers	Adidas						
14	08-DEC-DE	Coffee Maker	Nespresso						
15	18-FEB-CA	Smartwatch	Fitbit						
16	16-APR-ES	headphones	Bose						
17	21-AUG-CA	Laptop Bag	Samsonite						
18	20-AUG-CN	Smartwatch	Huawei						
19	27-JAN-IT	laptop	Asus	$980.00	10	Electronics	980.00	Electronics	
20	01-MAR-UK	Sunglasses	Oakley	$150.00	15	Fashion	150.00	Fashion	
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor	297.86	Outdoor	

4. Splitting and Merging Data:

● Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.



	A	B	C	D	E	F	G	H
1	Product II	Column1	Column2	Price ($)	Quantity	Category	Missing Pric	Missing Category
2	28	JAN	US	$1,000.00	30	Electronics	1000	Electronics
3	15	FEB	US	$80.00	15	Fashion	80	Fashion
4	3	MAR	US	$130.00	40	Kitchen	130	Kitchen
5	11	APR	US	$900.00	25	Electronics	900	Electronics
6	22	MAY	US	$70.00	20	Accessories	70	Accessories
7	7	JUN	UK		45	Electronics	309.3548387	Electronics
8	19	JUL	UK	$30.00	5	Fashion	30	Fashion
9	23	AUG	UK	$90.00	35	Kitchen	90	Kitchen
10	5	SEP	UK	$500.00	50	Electronics	500	Electronics
11	14	OCT	UK	$130.00	10	Outdoor	130	Outdoor
12	17	JUN	IN	$950.00	25	Electronics	950	Electronics
13	25	NOV	AU	$90.00	40		90	0
14	8	DEC	DE	$120.00	35		120	0
15	18	FEB	CA	$150.00	15	Electronics	150	Fashion
16	16	APR	ES	$250.00	20	Electronics	250	Electronics
17	21	AUG	CA	$50.00	35	Accessories	50	Kitchen
18	20	AUG	CN	$160.00	15	Electronics	160	Kitchen
19	27	JAN	IT	$980.00	10	Electronics	980	Electronics
20	1	MAR	UK	$150.00	15	Fashion	150	Kitchen
21	14	AUG	US		10	Outdoor	309.3548387	Kitchen



	A	B	C	D	E	F	G	H
1	Product II	Product Nam	Brand Nam	Price ($)	Quantity	Category	Missing Pric	Missing Category
2	28-JAN-US	laptop	Dell					
3	15-FEB-US	Sneakers	Nike					
4	03-MAR-US	Coffee Maker	Keurig					
5	11-APR-US	smartphone	Samsung					
6	22-MAY-US	Backpack	North Face					
7	07-JUN-UK	Headphones	Sony					
8	19-JUL-UK	T-shirt	Adidas					
9	23-AUG-UK	Blender	Ninja					
10	05-SEP-UK	Tablet	Apple					
11	14-OCT-UK	Hiking Boots	Timberland					
12	17-JUN-IN	Laptop	HP					
13	25-NOV-AU	Sneakers	Adidas					
14	08-DEC-DE	Coffee Maker	Nespresso					
15	18-FEB-CA	Smartwatch	Fitbit					
16	16-APR-ES	headphones	Bose					
17	21-AUG-CA	Laptop Bag	Samsonite					
18	20-AUG-CN	Smartwatch	Huawei					
19	27-JAN-IT	laptop	Asus					
20	01-MAR-UK	Sunglasses	Oakley					
21	14-AUG-US	Camping Tent	Coleman					

● Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

Product II	Column1	Cuntry Cod	Price ($)	Quantity	Category	Missing Pric	Missing Category
28-Jan-1900	JAN	US	$1,000.00	30	Electronics	1000	Electronics
15-Jan-1900	FEB	US	$80.00	15	Fashion	80	Fashion
3-Jan-1900	MAR	US	$130.00	40	Kitchen	130	Kitchen
11-Jan-1900	APR	US	$900.00	25	Electronics	900	Electronics
22-Jan-1900	MAY	US	$70.00	20	Accessories	70	Accessories
7-Jan-1900	JUN	UK		45	Electronics	309.3548387	Electronics
19-Jan-1900	JUL	UK	$30.00	5	Fashion	30	Fashion
23-Jan-1900	AUG	UK	$90.00	35	Kitchen	90	Kitchen
5-Jan-1900	SEP	UK	$500.00	50	Electronics	500	Electronics
14-Jan-1900	OCT	UK	$130.00	10	Outdoor	130	Outdoor
17-Jan-1900	JUN	IN	$950.00	25	Electronics	950	Electronics
25-Jan-1900	NOV	AU	$90.00	40		90	0
8-Jan-1900	DEC	DE	$120.00	35		120	0
18-Jan-1900	FEB	CA	$150.00	15	Electronics	150	Fashion
16-Jan-1900	APR	ES	$250.00	20	Electronics	250	Electronics
21-Jan-1900	AUG	CA	$50.00	35	Accessories	50	Kitchen
20-Jan-1900	AUG	CN	$160.00	15	Electronics	160	Kitchen
27-Jan-1900	JAN	IT	$980.00	10	Electronics	980	Electronics
1-Jan-1900	MAR	UK	$150.00	15	Fashion	150	Kitchen
14-Jan-1900	AUG	US		10	Outdoor	309.3548387	Kitchen

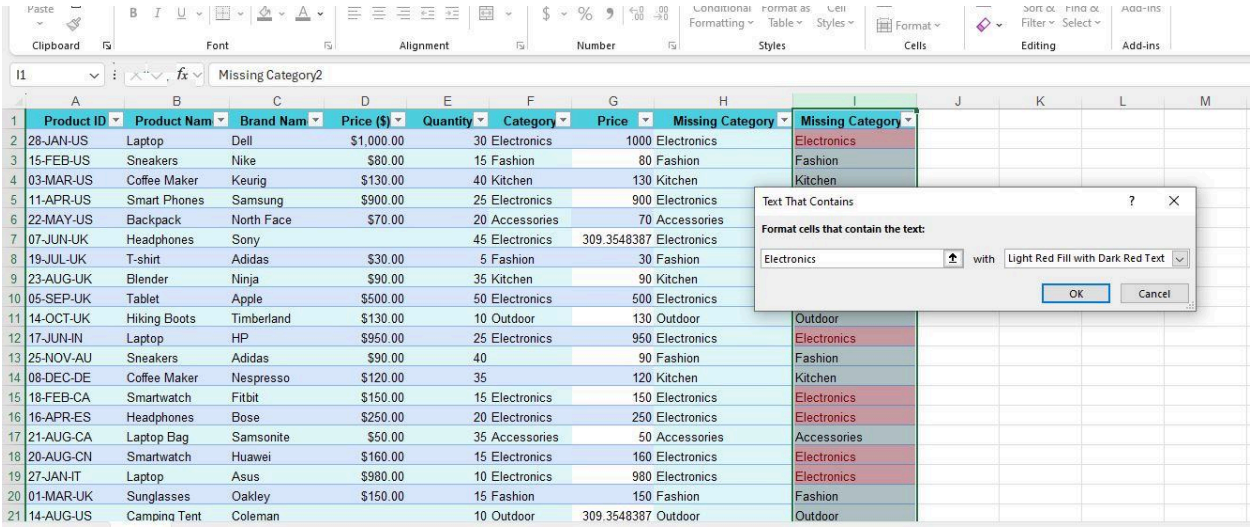
6. Conditional Formatting:

● Apply data bar or color scales conditional formatting in the "Price" column.



	A	B	C	D	E	F	G	H	I
	Product Il	Column1	Cuntry Cod	Price ($)	Quantity	Category	Missing Pric	Missing Category	Manufactu
2	28-Jan-1900	JAN	US	$1,000.00	30	Electronics	1000	Electronics	
3	15-Jan-1900	FEB	US	$80.00	15	Fashion	80	Fashion	
4	3-Jan-1900	MAR	US	$130.00	40	Kitchen	130	Kitchen	
5	11-Jan-1900	APR	US	$900.00	25	Electronics	900	Electronics	
6	22-Jan-1900	MAY	US	$70.00	20	Accessories	70	Accessories	
7	7-Jan-1900	JUN	UK		45	Electronics	309.3548387	Electronics	
8	19-Jan-1900	JUL	UK	$30.00	5	Fashion	30	Fashion	
9	23-Jan-1900	AUG	UK	$90.00	35	Kitchen	90	Kitchen	
10	5-Jan-1900	SEP	UK	$500.00	50	Electronics	500	Electronics	
11	14-Jan-1900	OCT	UK	$130.00	10	Outdoor	130	Outdoor	
12	17-Jan-1900	JUN	IN	$950.00	25	Electronics	950	Electronics	
13	25-Jan-1900	NOV	AU	$90.00	40		90	0	
14	8-Jan-1900	DEC	DE	$120.00	35		120	0	
15	18-Jan-1900	FEB	CA	$150.00	15	Electronics	150	Fashion	
16	16-Jan-1900	APR	ES	$250.00	20	Electronics	250	Electronics	
17	21-Jan-1900	AUG	CA	$50.00	35	Accessories	50	Kitchen	
18	20-Jan-1900	AUG	CN	$160.00	15	Electronics	160	Kitchen	
19	27-Jan-1900	JAN	IT	$980.00	10	Electronics	980	Electronics	
20	1-Jan-1900	MAR	UK	$150.00	15	Fashion	150	Kitchen	
21	14-Jan-1900	AUG	US		10	Outdoor	309.3548387	Kitchen	

● Create a
custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."



	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price	Missing Category	Missing Category
2	28-JAN-US	Laptop	Dell	$1,000.00	30	Electronics	1000	Electronics	Electronics
3	15-FEB-US	Sneakers	Nike	$80.00	15	Fashion	80	Fashion	Fashion
4	03-MAR-US	Coffee Maker	Keurig	$130.00	40	Kitchen	130	Kitchen	Kitchen
5	11-APR-US	Smart Phones	Samsung	$900.00	25	Electronics	900	Electronics	Electronics
6	22-MAY-US	Backpack	North Face	$70.00	20	Accessories	70	Accessories	Accessories
7	07-JUN-UK	Headphones	Sony		45	Electronics	309.3548387	Electronics	Electronics
8	19-JUL-UK	T-shirt	Adidas	$30.00	5	Fashion	30	Fashion	Fashion
9	23-AUG-UK	Blender	Ninja	$90.00	35	Kitchen	90	Kitchen	Kitchen
10	05-SEP-UK	Tablet	Apple	$500.00	50	Electronics	500	Electronics	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	$130.00	10	Outdoor	130	Outdoor	Outdoor
12	17-JUN-IN	Laptop	HP	$950.00	25	Electronics	950	Electronics	Electronics
13	25-NOV-AU	Sneakers	Adidas	$90.00	40		90	Fashion	Fashion
14	08-DEC-DE	Coffee Maker	Nespresso	$120.00	35		120	Kitchen	Kitchen
15	18-FEB-CA	Smartwatch	Fitbit	$150.00	15	Electronics	150	Electronics	Electronics
16	16-APR-ES	Headphones	Bose	$250.00	20	Electronics	250	Electronics	Electronics
17	21-AUG-CA	Laptop Bag	Samsonite	$50.00	35	Accessories	50	Accessories	Accessories
18	20-AUG-CN	Smartwatch	Huawei	$160.00	15	Electronics	160	Electronics	Electronics
19	27-JAN-IT	Laptop	Asus	$980.00	10	Electronics	980	Electronics	Electronics
20	01-MAR-UK	Sunglasses	Oakley	$150.00	15	Fashion	150	Fashion	Fashion
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor	309.3548387	Outdoor	Outdoor